do280-4.18: Days remaining **89** | Lab hours used **4/80**

DO280-4.22: Days remaining **89** | Lab hours used **0/80**

Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Lab: Set up Cluster Self-service

Configure a cluster with default settings for self-service projects.

Outcomes

- Create a project template that sets quotas, ranges, and network policies.
- Restrict access to the `self-provisioners` cluster role.
- Create groups and assign users to groups.
- Use RBAC to grant permissions to groups.

NOTE

The allocated time for this activity is 60 minutes. If you need additional time to complete the task, then you must revisit the course content, or practice more.

As the `student` user on the `workstation` machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available:

```
[student@workstation ~]$ lab start compreview-review
```

The `lab` command creates the following users:

User	Password
`do280-support`	`redhat`
`do280-platform`	`redhat`
`do280-presenter`	`redhat`
`do280-attendee`	`redhat`

Specifications

A leading IT solutions provider aims to host workshops on various topics. The company requires a dedicated OpenShift cluster to manage multiple workshops. Each workshop must be isolated to prevent interference between them.

You are the OpenShift administrator and need to configure the cluster so that each workshop gets a separate project.

When a presenter creates a project for a workshop, the project is provisioned with a base configuration that consists of resource quotas, RBAC, limit ranges, and network policies. The presenter must be able to manage workshop-related resources with minimal intervention from the support team.

The workshop support team ensures smooth operation, and intervenes only when technical issues arise. The workshop support team must deploy applications that administer workshops and that enhance the workshop experience. You set up a project and the applications for this purpose on a second lab.

This setup enables scalability, security, and ease of administration for workshop environments.

Because this exercise requires steps that restart the Kubernetes API server, this configuration must persist across API server restarts.

The cluster API URL is `https://api.lab.example.com:6443`. The password for the `developer` user is `developer`, and the password for the `admin` user is `redhatocp`.

> **NOTE**
>
> All the resources that the cluster creates with a new workshop project must use `workshop` as the name for grading purposes. The `lab start` command provides incomplete resource definitions in the `~/DO0019L/labs/compreview-review` directory.

directory and solution files in the `~/DO0019L/solutions/compreview-review` directory.

Configure the cluster with the following groups and roles. Ensure that only the users for these groups can create projects, and that the configuration persists across API server restarts.

Group	User	Description
`platform`	`do280-platform`	Group that is responsible for cluster administration
`presenters`	`do280-presenter`	Group that is responsible for delivering the workshop
`workshop-support`	`do280-support`	Group that is responsible for workshop administration

Configure a project template so that each workshop is hosted in an independent project. The template must provision the following resources for each new project:

Verify the configuration by creating a workshop and setting up the workshop support team.

- The `workshop-support` group has the `admin` cluster role.
- The `workshop-support` group has the `manage-groups` custom cluster role. The role grants support team members the right to create and manage workshop groups and to add workshop attendees. The custom role allows all the operations in the `groups` resource.
- The `platform` group must be able to administer the cluster without restrictions.
- The project template must create a role binding that is named `workshop` and that grants the `admin` cluster role to the project creator in the workshop's namespace.
- The workshop's project must have the `workshop=`*`project_name`* label to help identify the workshop workload.
- Enforce the following maximum quota for each workshop:
  - The project uses up to 2 CPUs.
  - The project uses up to 1 GiB of RAM.
  - The project requests up to 1.5 CPUs.
  - The project requests up to 750 MiB of RAM.
- Enforce the following limits to prevent an attendee's workload from consuming all the allocated resources for the workshop:
  - A workload uses up to 750m CPUs.
  - A workload uses up to 750 MiB of memory.
- Have the following resource limits for workloads:

  - A default limit of 500m CPUs
  - A default limit of 500 MiB of RAM
  - A default request of 0.1 CPUs
  - A default request of 250 MiB of RAM
- The pods that are deployed by the `workshop-support` team must accept traffic only from pods within the same workshop project (which is identified by the `workshop=`*`project_name`* label) or from the ingress router (which is identified by the `policy-group.network.openshift.io/ingress: ""` namespace label).
- The `workshop-support` group uses the `registry.lab.example.com:8443/redhattraining/hello-world-nginx:v1.0` image, which listens on the `8080` port, to simulate a workshop workload.
- As the `do280-presenter` user, you must create a workshop with the `do280` name.
- The `do280-support` user must perform the following tasks for the workshop project:
  - Create the `do280-attendees` group.
  - Add the `do280-attendee` user to the `do280-attendees` group.
  - Assign the `edit` role to the `do280-attendees` group in the `do280` project.
- The `do280-attendee` user can create workloads in the `do280` project.
- The `do280-attendee` user cannot create projects.

> **NOTE**
>
> The `lab` command provides incomplete resource definitions in the `~/DO0019L/labs/compreview-review` directory. The `lab` command also copies the solution files for the exercise to the `~/DO0019L/solutions/compreview-review` directory.

1. Create the groups and add the users as specified in the following table:

Group	User
`platform`	`do280-platform`
`presenters`	`do280-presenter`
`workshop-support`	`do280-support`

2. Grant privileges to groups by using cluster roles and cluster role bindings as specified in the following table. You can use the `~/DO0019L/labs/compreview-review/groups-role.yaml` file to create the `manage-groups` role.

Role	Group
admin	workshop-support
manage-groups	workshop-support
cluster-admin	platform

3. Allow only the `platform`, `workshop-support`, and `presenters` groups to create projects.

   3.1. Edit the `self-provisioners` cluster role binding to enforce that only users from the `platform`, `workshop-support`, and `presenters` groups can create projects.

   3.2. Make this change permanent by setting the `rbac.authorization.kubernetes.io/autoupdate` annotation to `false`.

   3.3. Verify that you cannot create a project as the `do280-attendee` user.

4. Create the project template with the quota, limit range, and network policy.

   4.1. As the `admin` user, create a `template-test` namespace to design the project template.

   4.2. Create the quota and the limit range in the `template-test` namespace. You can use the `~/DO0019L/labs/compreview-review/quota.yaml` and `~/DO0019L/labs/compreview-review/limitrange.yaml` files to create the quota and the limit range.

   4.3. Create a network policy to accept traffic from within the workshop project or from outside the cluster. To identify the workshop project traffic, label the `template-test` namespace with the `workshop=template-test` label. You can use the `~/DO0019L/labs/compreview-review/networkpolicy.yaml` file to create the network policy.

   4.4. Create the workshop project template by using the previously created template resources. You can use the `oc adm create-bootstrap-project-template -o yaml > template.yaml` command to create the template for new projects.

5. As the `do280-presenter` user, create the `do280` workshop project and confirm the project template resources.

   5.1. Verify that the `do280` project has the quota, limit range, and network policy that are specified in the project template that you configured.

   5.2. Confirm that the `do280` project definition has the `workshop=do280` label.

   5.3. Verify that you can connect to the `workshop` pod only from the `workshop` project.

6. As the `do280-support` user, create the `do280-attendees` group, grant privileges, and confirm that the `do280-attendee` user can create workloads.

   6.1. Assign the `edit` role to the `do280-attendees` group, and add the `do280-attendee` user to the group.

   6.2. Log in as the `do280-attendee` user, and verify that you can create workloads in the `do280` project.

Hide Solution

Evaluation

Evaluate the allocated time for this activity against the time that you took to complete it. If you used more time than was allocated, then you must revisit the course content, or practice more.

> **NOTE**
>
> The grading scripts provide remediation hints to help you identify what content you must revisit to complete the tasks.

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade compreview-review
```

Finish

As the `student` user on the `workstation` machine, use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish compreview-review
```

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